



**NEW YORK INSTITUTE
OF TECHNOLOGY**

Rheumatologic Diseases Affecting the Joints

Principles and Practice of Osteopathic Medicine I

Recorded Spring 2026

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Session Objectives

At the conclusion of the session/course, the student should be able to:

Describe the presentation and diagnostic tests used to diagnose the following conditions:

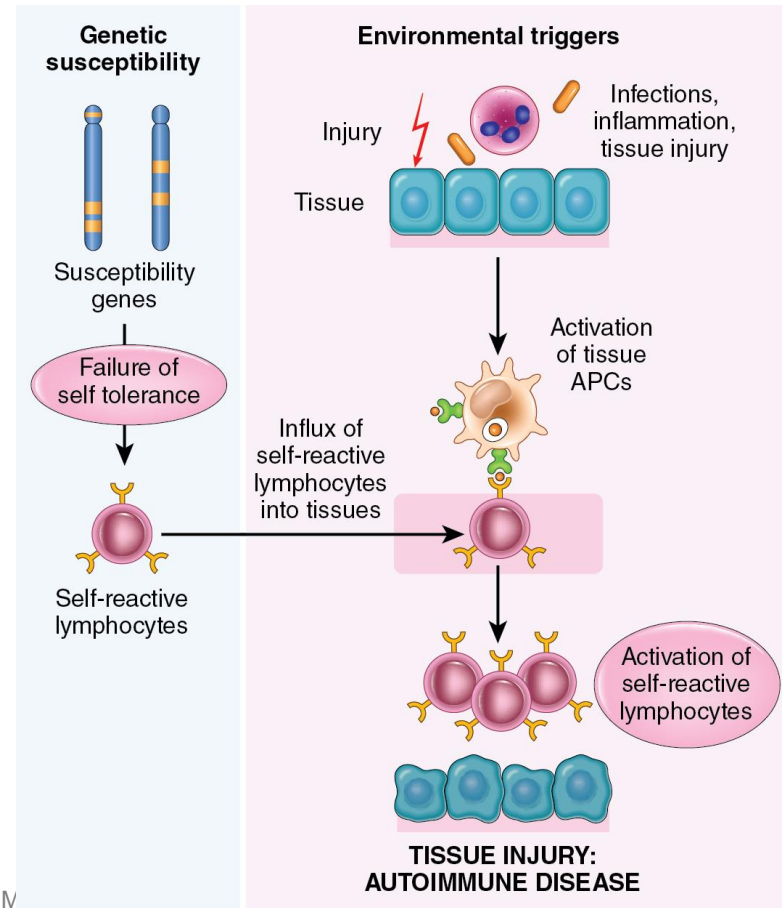
- Rheumatoid Arthritis
- Osteoarthritis
- Gout
- Pseudogout

Describe seronegative spondyloarthritis presentations including: ankylosing spondylitis, psoriatic arthritis, and reactive arthritis.

Differentiate between causes of conditions affecting primarily the joints

Overview of Autoimmune disease

- Genetic Predisposition
 - Association with HLA Genes
- Environmental triggers
- Tissue Injury



Jon C. Aster, MD, PhD; Abul K. Abbas, MBBS; Vinay Kumar, M
167-235

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Who, What, When, Where, Why, How

Questions to ask as you go through this lecture:

- **What** is it? How does it present, how do you diagnose it?
- **Who** gets it? It occur in certain demographics?
- **When** does it occur? Certain age range?
- **Where** does it occur? Certain joints?
- **How** do we diagnose it? Presentation
- **Why** does it matter? Quality of life?

A 45-year-old woman with joint pain...

A 45 y/o woman presents to her primary care doctor with hand pain and stiffness affecting both hands for the past 6 months...



- Chronic
- Principally attacks joints
- Nonsuppurative, proliferative, inflammatory synovitis.
- Autoimmune response: CD4+ helper T cells reacting against a joint antigen.
- Associations:
 - HLA-DR4
 - Smoking
 - Infection

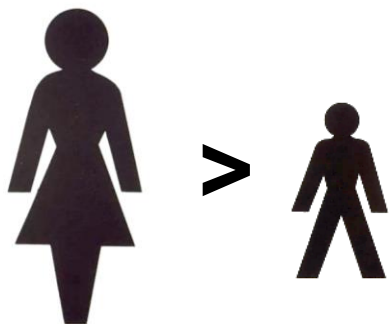
Rheumatoid Arthritis Presentation

What

- Joint pain and stiffness
- Joint swelling and redness
- Symptoms affecting multiple joints including the hands and fingers
- Symmetrical symptoms
- Morning stiffness lasting about 30-60 minutes
 - Or after prolonged inactivity
- Symptoms lasting longer than six consecutive months
- Extra-articular symptoms
 - Malaise, fatigue.

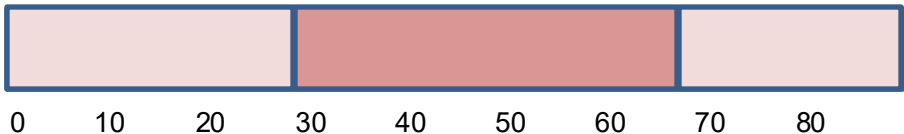
Who

Epidemiology of Rheumatoid Arthritis



Sex:
3:1 female:male

Associated with
Smoking use



Can occur at any age, Peaks 30-40s

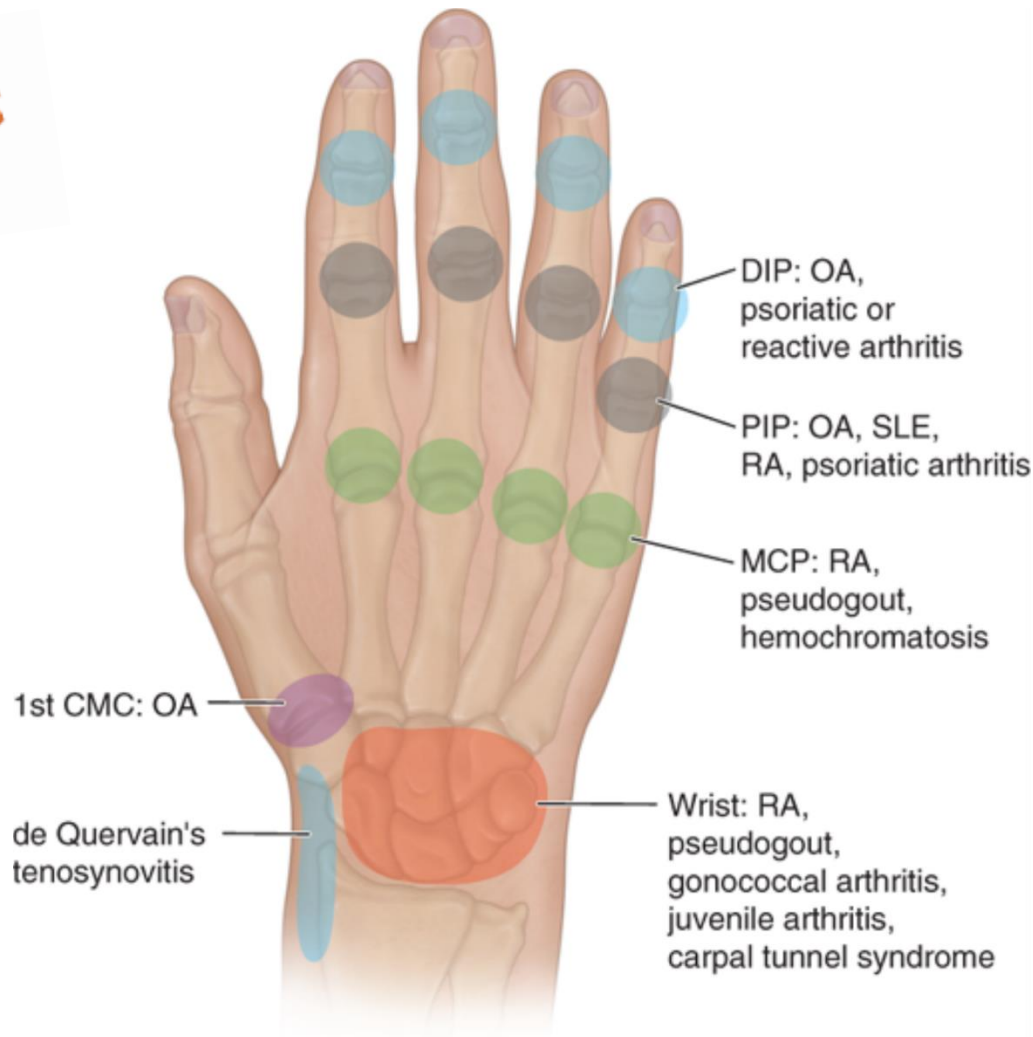
When



Affects about 1% of the US
population

Overview of conditions affecting the hand and wrist...

Where



Extra-Articular Symptoms/Signs

- Fever, fatigue, weight loss
- Felty Syndrome: RA, Neutropenia, and Splenomegaly
- Rheumatoid nodules



Source: Maxine A. Papadakis, Stephen J. McPhee, Michael W. Rabow:
Current Medical Diagnosis and Treatment 2020
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Rheumatoid arthritis with ulnar deviation at the metacarpophalangeal (MCP) joints. (Used, with permission, from Richard P. Usatine, MD.)



Source: Maxine A. Papadakis, Stephen J. McPhee, Michael W. Rabow:
Current Medical Diagnosis and Treatment 2020
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Rheumatoid nodules over the extensor surface of the forearm. (Used, with permission, from Richard P. Usatine, MD, in Usatine RP, Smith MA, Mayeaux EJ Jr, Chumley H. The Color Atlas of Family Medicine, 2nd ed. McGraw-Hill, 2013.)

Classic findings of the hands

Swan Neck Deformity



Boutonniere deformity



Joints can have erosions, juxta-articular osteopenia, joint space narrowing, ulnar finger deviation

- Base diagnosis on clinical symptoms and findings...
- Use and radiographic findings to support the diagnosis and rule out similar diseases
- Key labs:
 - Rheumatoid Factor (sensitive but not specific)
 - Anti-Citrullinated peptide Ab (highly specific ~95%)
 - ANA (~30-60% sensitivity)
 - ESR or CRP can help assess severity of dz

Overview of tests (sensitivity of tests)

	RA	SLE	Sjögren	Diffuse Scleroderma	Limited scleroderma (CREST syndrome)	Polymyositis/ dermatomyositis
ANA	30–60	95–100	95	>95	>95	80
Anti-ds DNA	0–5	60	0	0	0	0
Rheumatoid Factor	70	20	75	30	30	33
Anti-Sm	0	10–25	0	0	0	0
Anti-SS-A	0–5	15–20	65	0	0	0
Anti-SS-B	0–2	5–20	65	0	0	0
Anti-SCL-70	0	0	0	33	20	0
Anti-Centromere	0	0	0	1	50	0
Anti-Jo-1	0	0	0	0	0	20–30
ANCA	0	0–1	0	0	0	0

Radiographic Findings of RA



Joints can have erosions, juxta-articular osteopenia, joint space narrowing, ulnar finger deviation, swan neck deformity

Source: Radiopedia – link: <https://radiopaedia.org/cases/rheumatoid-arthritis-13>

RA Treatment

Treat:

- Symptoms, inflammation, and progression

Use DMARDs:

- Corticosteroids, immunosuppressants, methotrexate, biologics: TNF α inhibitors.

- Goals of treatment:
 - Minimize joint pain and swelling,
 - Prevent joint deformity
 - Maintain quality of life
 - Control extra-articular manifestations
 - Both caused by disease and treatment
- Prognosis – dependent on severity however ~12-15 year decrease of life expectancy related to CV disease

A 55 year-old man with joint pain...

A 55 y/o man presents to his primary care doctor with hip pain affecting his left hip for the past 10 years but now limiting his daily activities...



Osteoarthritis

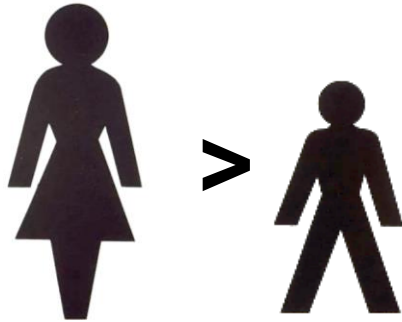
- Degenerative disease of Articular Cartilage
- Secondary Inflammation
- Association:
 - Wear-and-tear
 - Predisposed deformity.

What

Who

Epidemiology of Osteoarthritis

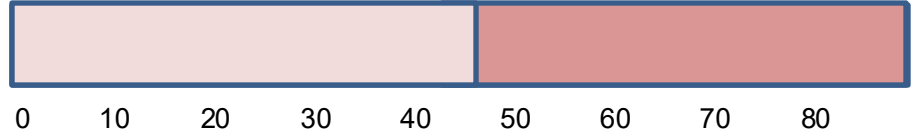
When



Sex:
More Common in
Women than Men



No conclusive
evidence of genetic
relationship



Typically begins in late 40s
with increasing as older



Obesity, competitive contact
athletics, and occupational risks
are positive risk factors

Classic Findings in Osteoarthritis

What

- Insidious onset is often found
- Pain initially is short in duration and later prolonged
- Pain increases with activity or stress
- Joint deformities may further develop and might inhibit activity
- Joints affected are often related to overuse or prior injury (ie, fracture site)



Source: Maxine A. Papadakis, Stephen J. McPhee, Michael W. Rabow:
Current Medical Diagnosis and Treatment 2020
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Osteoarthritis in an older woman with **Heberden nodes** at the distal interphalangeal joints. There is some swelling beginning at the proximal interphalangeal joints creating **Bouchard nodes**. (Used, with permission, from Richard P. Usatine, MD.)



Source: Maxine A. Papadakis, Stephen J. McPhee, Michael W. Rabow:
Current Medical Diagnosis and Treatment 2020
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Osteoarthritis with medial compartment narrowing of both knees. (Used, with permission, from Nicole Richman, MD.)

- History and physical
 - Lack of extraarticular manifestations
- Xray findings are indicative
- Laboratory findings will be negative
 - ESR negative
 - Aspiration of synovial fluid should be normal

Treatment Guidelines for OA

HAND

KNEE

HIP

Exercise*

Self-Efficacy and Self-Management Programs

Weight Loss

Tai Chi

Cane

1st CMC Orthosis

TF Knee Brace**

Heat, Therapeutic Cooling

Cognitive Behavioral Therapy

Acupuncture

Kinesiotaping

Balance Training

Other Hand Orthoses***

PF Knee Brace**

Paraffin

Yoga

RFA

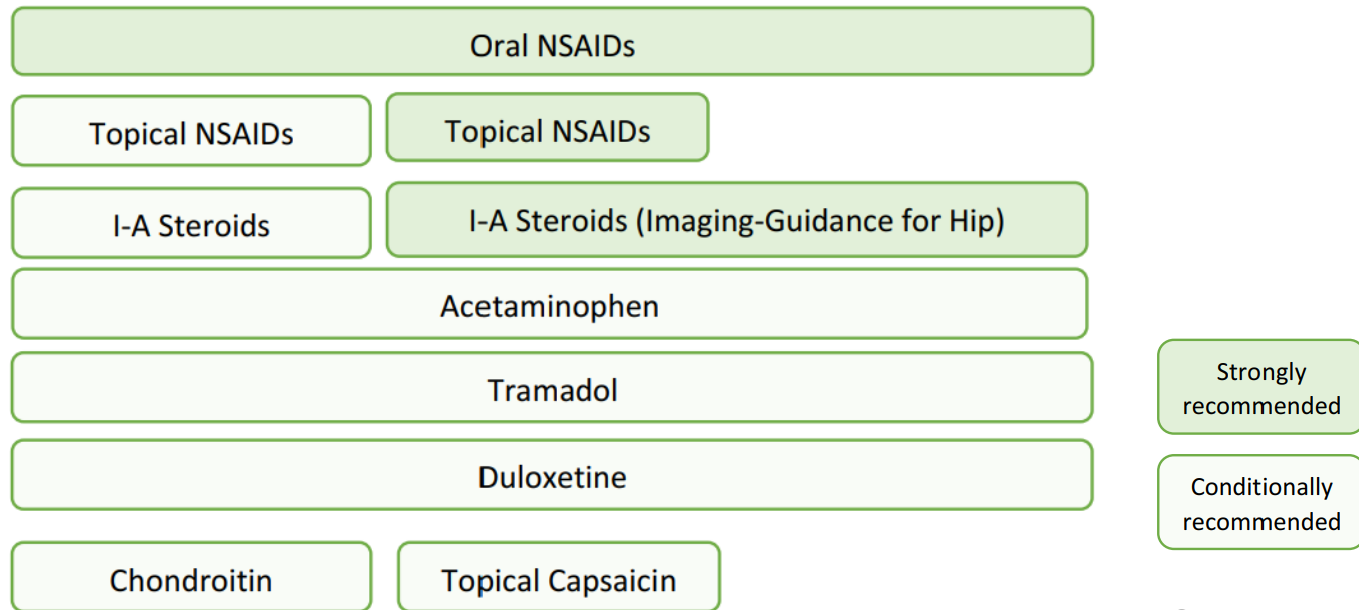
Strongly
recommended

Conditionally
recommended

Source: 2019 American College of Rheumatology Guideline for the Treatment of Osteoarthritis

Treatment Guidelines for OA

PHARMACOLOGIC APPROACHES



Source: 2019 American College of Rheumatology Guideline for the Treatment of Osteoarthritis

A 65 year-old man with joint pain...

A 65 y/o man presents to the urgent care with severe pain in his right first toe that started last night...

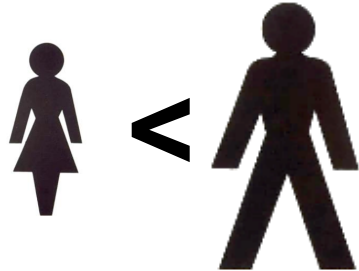


- Transient Attacks
- Monosodium urate crystals deposits within and around joints.
- Primary cause
 - Enzyme defects
- Secondary causes
 - Renal disease, congenital, leukemia
- Excessive Uric Acid
 - majority is **reduced excretion**
 - Increased rate of synthesis

Who

Epidemiology of Gout

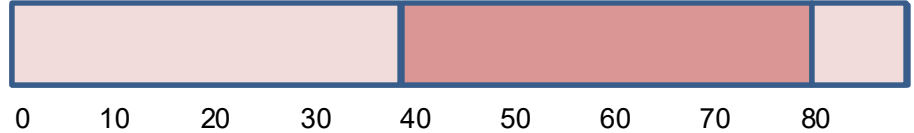
When



Sex:
Female < Male



Possible genetic
role related to
purine metabolism



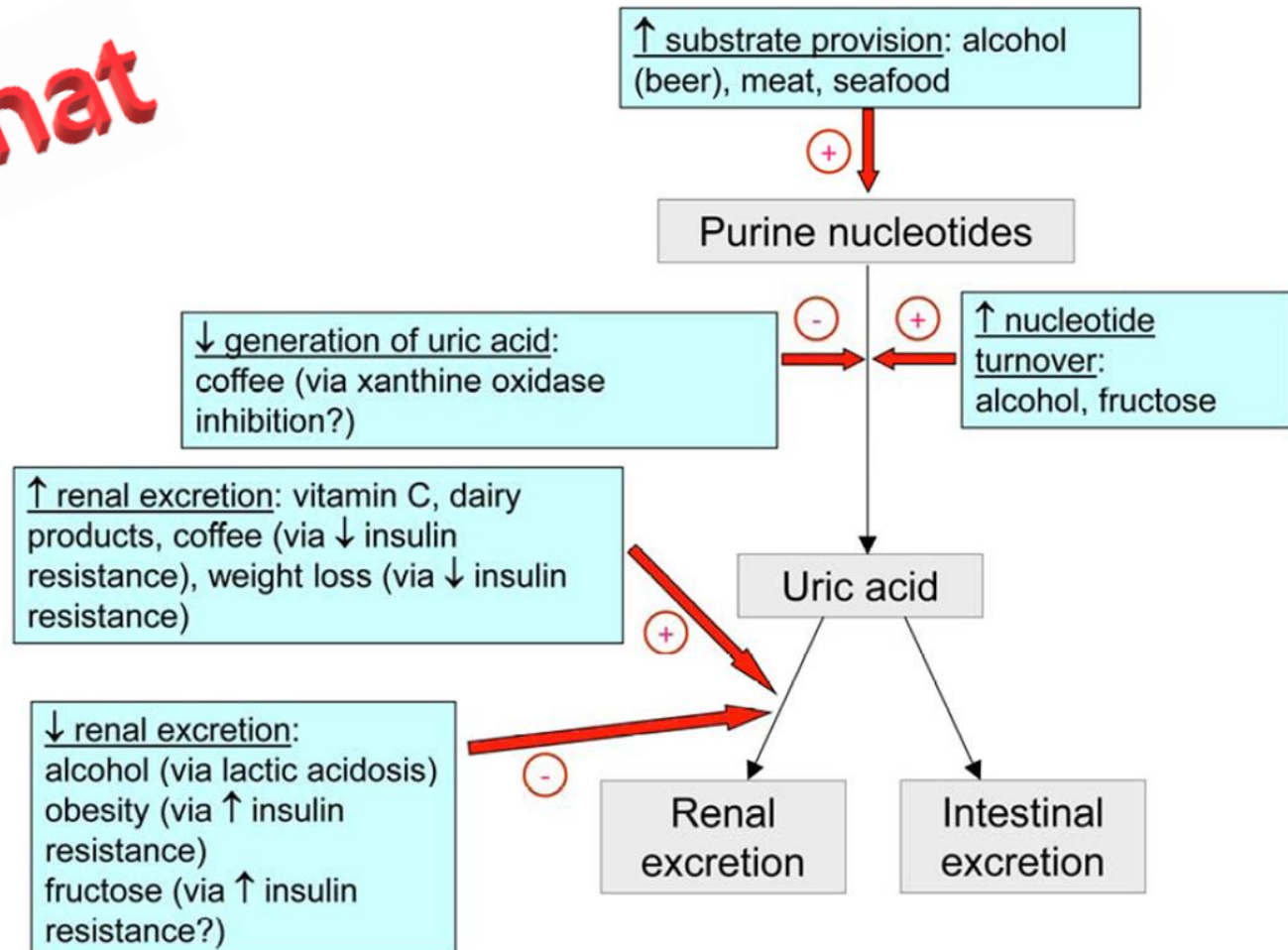
Can occur at any age however
increased incidence starting in 40s
with a peak at 75.



Variable affect globally related to
diet (higher socioeconomic) and
background (Pacific Islanders)

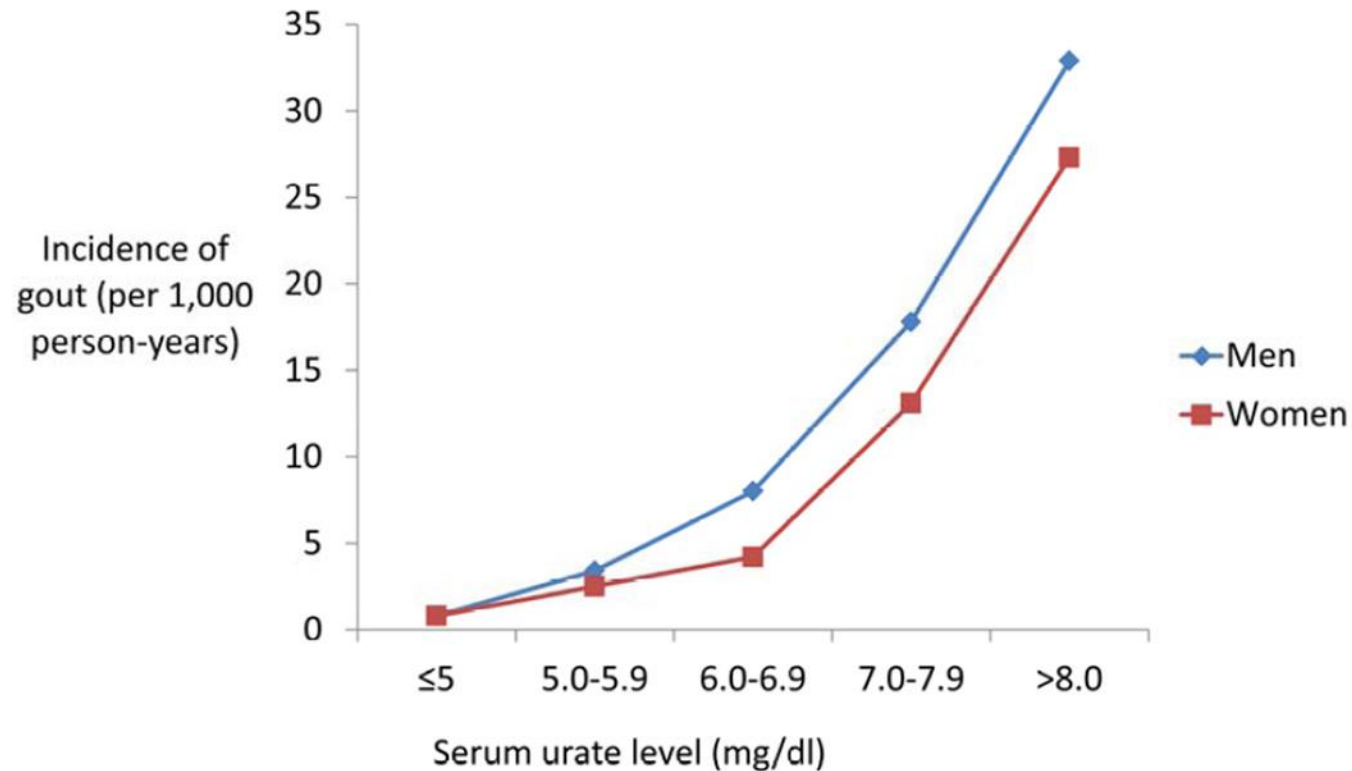
Diet and Gout

What



Source: Roddy, Edward, and Hyon K Choi. "Epidemiology of gout." Rheumatic diseases clinics of North America vol. 40,2 (2014): 155-75. doi:10.1016/j.rdc.2014.01.001

Relationship of gout and uric acid



Source: Bhole V, de Vera M, Rahman MM et al. Epidemiology of gout in women: fifty-two-year followup of a prospective cohort. Arthritis Rheum 2010;62:1069-1076.

Classic Presentation of Gout

- Male patient
- Obese
 - or Recent weight gain
- Consumption of EtOH
- Diet high in meat/seafood
- Thiazides
- Renal impairment hx
- Painful joint
- Erythematous
- Swollen
- Warm joint
- Typically monoarticular

Gout Tophi



Source: <http://www.pcds.org.uk/clinical-guidance/gouty-tophi>

Due to repetitive precipitation of urate crystals.

Making the diagnosis of Gout

How

- Clinical suspicion
- Joint Aspiration
 - **Negative birefringent crystals**
- Serum Uric Acid $\uparrow$
 - 20% of people with high Uric Acid will develop gout



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: Harrison's Principles of Internal Medicine, 20th Edition
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Treatment of Gout

Anti-inflammatory

NSAIDS

Colchicine

Glucocorticoids

Urate Lowering Drugs

Uricosuric agents
(probenecid)

Xanthine
Oxidase Inhib
(allopurinol)

Lifestyle

Decrease purine
rich foods

Decreases/stop
EtOh

Weight loss

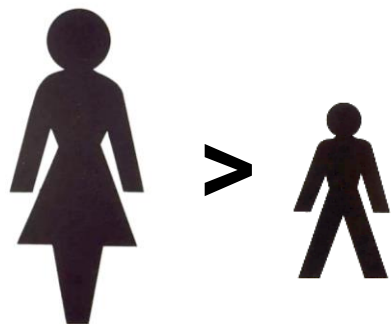
A 65 year-old woman with joint pain...

A 65 y/o woman presents to the urgent care with severe pain in her right knee that started last night...



Who

Epidemiology of CPPD Pseudogout



Sex:
Female > Male



Potential genetic
relationship to
pyrophosphate production



Majority of cases after age 50

When

Can be:

Sporadic
Hereditary
secondary

Gout Vs. Pseudogout

- Calcium Pyrophosphate deposition disease
 - Formerly known as Pseudogout
- Crystals form in hyaline cartilage and fibrocartilage.
 - develop in articular cartilage, menisci, and intervertebral discs.
- *Knee: Think pseudogout, not Gout*
- *Toe: think gout*

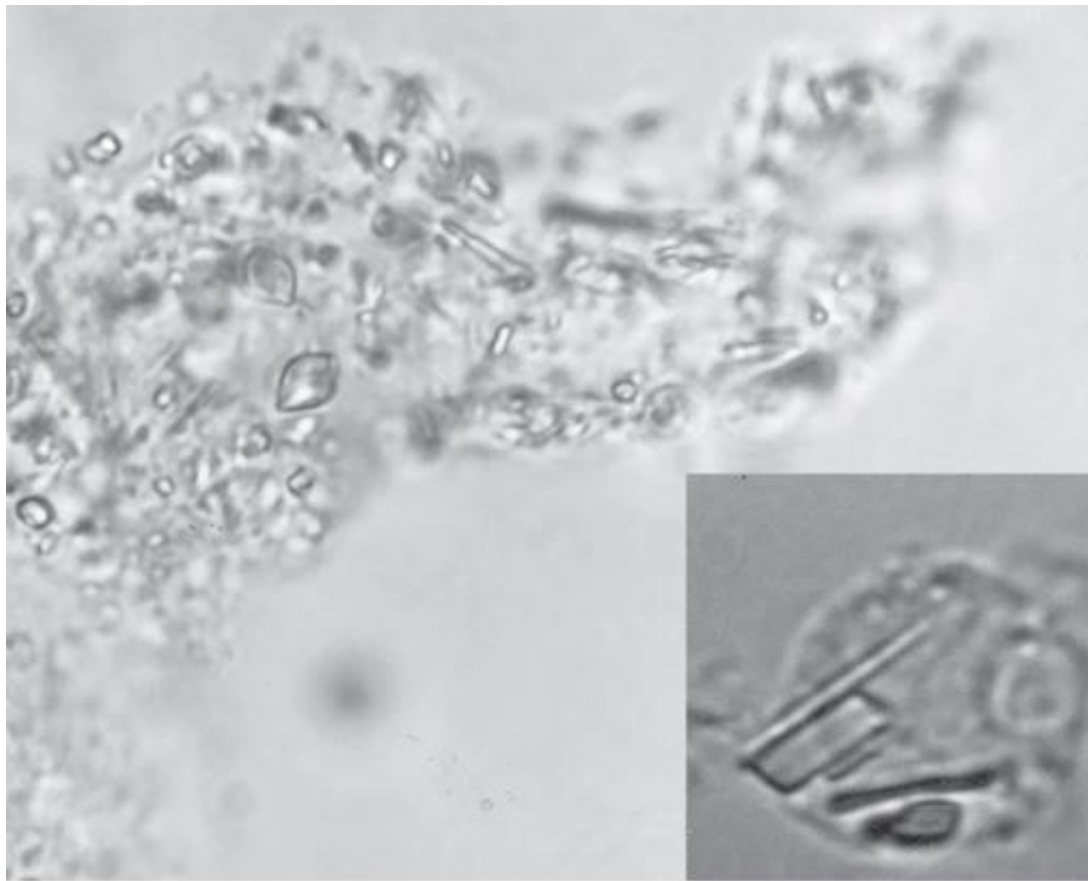
Where

Making the diagnosis of Pseudogout

- Clinical history
 - Typically, inclusive of a prior injury of the joint
 - Recent reinjury (could be minor)
- Warm, painful joint with fever
- Aspiration of joint with weakly positive birefringent crystals

What

How



Source: J.L. Jameson, A.S. Fauci, D.L. Kasper, S.L. Hauser, D.L. Longo, J. Loscalzo: *Harrison's Principles of Internal Medicine*, 20th Edition
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Intracellular and extracellular calcium pyrophosphate (CPP) crystals, as seen in a fresh preparation of synovial fluid, illustrate rectangular, rod-shaped, and rhomboid crystals that are weakly positively or nonbirefringent crystals (compensated polarized light microscopy; 400 ×).

Treatment of Pseudogout

- Rest
- Oral or injectable NSAIDs
- Intra-articular glucocorticoid injection
- Recurrent events – consider colchicine prophylaxis
- Surgery if joint destruction/deformity occurs

Seronegative Spondyloarthritis

- Associated with HLA-B27
- Absent Rheumatoid factor.
- Probable T-Cell response.
- Affects ligamentous attachments instead of synovium
- Involves SI and vertebral joints.
- Bony proliferation leading to ankylosis.
- Includes:
 - *Ankylosing spondylitis, Psoriatic Arthritis, Reactive arthritis, and association with Inflammatory Bowel Disease*

Who

Ankylosing Spondylitis



Sex: Historically:
Male > Female



90% HLA-B27 Positive

Starts Second and Third Decade

When

Presentation: lower back pain and
spinal immobility.

1/3 Cases affect peripheral joints.

Psoriatic Arthritis

Who



Sex:
Male = Female



HLA-B27 Positive

When

Ages 30-55
History of Psoriasis

Presentation:

Dactylitis, arthritis, spondylitis,
enthesitis.

Asymmetric multiple or one joint

X-ray: pencil in cup deformity
(see *Scholar-Rx brick*)

Reactive Arthritis



Globally:
30/100,000 adults

Who

Assoc. with following infection:
Chlamydia, Shigella, Salmonella, Yersinia, Campylobacter, Clostridioides.

When



HLA-B27

Presentation:

Arthritis

Urethritis/Cervicitis

Conjunctivitis

50% recurrent joint pains

Key words to define

- HLA
- Birefringent
- ANA
- Rheumatoid Factor
- Anti-CCP
- ESR
- Synovitis
- Dactylitis
- Spondylitis
- Enthesitis
- Ankylosis

References

Main Reference

Robbins, Cotran & Kumar. Pathologic Basis of Disease. Jon C. Aster, et al. Amsterdam: Elsevier, 2025.

Ch. 26: <https://nyit.idm.oclc.org/login?url=https://www.clinicalkey.com/#!/content/book/3-s2.0-B9780443264528000357#h10002470>

Older References:

- Rheumatoid Arthritis. Current Medical Diagnosis and Treatment 2020. Section 20-05. Maxine A. Papadakis, et al. New York, NY: McGraw-Hill. 2020. Link: https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?sectionid=279931_084&bookid=3343&Resultclick=2
- Degenerative Arthritis (Osteoarthritis). Current Medical Diagnosis and Treatment 2020. Section 20-02. Maxine A. Papadakis, et al. New York, NY: McGraw-Hill. 2020. Link: https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?sectionid=26913_3742&bookid=3212#269133761
- Gout, Pseudogout, and Related Diseases. In: Jameson J, Fauci AS, Kasper DL, Hauser SL, Longo DL, Loscalzo J. eds. Harrison's Manual of Medicine, 20e New York, NY: McGraw-Hill. 2020. Link: https://nyit.idm.oclc.org/login?url=https://accessmedicine.mhmedical.com/content.aspx?sectionid=227_559483&bookid=2738&Resultclick=2
- Supplementally - consider reviewing the ScholarRx Brick titled "Arthritis: Putting it All Together" https://usmle-rx.scholarrx.com/rx-bricks/brick/CP_MUS0004

Lecture Feedback Form:

<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>